

#HW 10

● Graded

26 Days, 23 Hours Late

Student

Vianey Gonzalez

Total Points

93 / 100 pts

Question 1

(no title)

30 / 30 pts

✓ - 0 pts Correct

Question 2

(no title)

6 / 10 pts

- 0 pts Correct

✓ - 2 pts Mass flow rate = 0.53 kg/s

- 0 pts Click here to replace this description.

✓ - 2 pts Temperature= -10 C

Question 3

(no title)

14 / 15 pts

- 0 pts Correct

- 1 pt You have to add the two to get the total power -406 kW.

- 1 pt Power -406.2 kW (calculation is wrong.)

✓ - 1 pt Power should be in negative.

- 1 pt Mass flowrate should be 2.2 kg/s.

- 6 pts incomplete

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt The answer will be positive.

- 6 pts incomplete

Question 5

(no title)

14 / 15 pts

– 0 pts Correct

✓ – 1 pt The unit of temperature is degrees C.

– 1 pt The temperature will be around 575 degree C.

– 1 pt Work will be around 15-17 MW

– 6 pts incomplete

Question 6

(no title)

14 / 15 pts

– 0 pts Correct

✓ – 1 pt The second answer should be around 1327 s.

– 6 pts incomplete

No questions assigned to the following page.

Thema

HW 10

Questions assigned to the following page: [1](#), [2](#), [3](#), [4](#), [5](#), and [6](#)

7.18.72

$$\dot{m} = \rho A v = (1000)(1500) = 1.5 \times 10^6 \text{ kg/s}$$

$$\dot{Q}_{cv} - \dot{W}_{cv} = \dot{m} \left(h_1 + \frac{V_1^2}{2} + gz_1 \right) - \dot{m} \left(h_2 + \frac{V_2^2}{2} + gz_2 \right)$$

$$\dot{Q}_{cv} - \dot{W}_{cv} = \dot{m} (gz_1 - gz_2) = (1.5 \times 10^6) (9.8) (20 \text{ m}) = 2.94 \times 10^7 \text{ W} = 29.4 \text{ MW}$$

turning over $29.43 - 0.06 = 28.8 \text{ MW}$

Electrical power generated during peak operation < pump power input

7.18.71

$$\dot{W}_A = \dot{m} c_p (T_2 - T_1) = 0.6(1.004)(300) = 181 \text{ kW}$$

$$\dot{W}_B = \dot{m} c_p (T_2 - T_1) = 0.6(1.004)(350) = 211 \text{ kW}$$

$$\sum \dot{m} h_e - \sum \dot{m} h_i = 0$$

$$\dot{m} c_p (h_2 - h_1) + \dot{m} c_p (h_3 - h_2) = 0 \rightarrow SS$$

$$\dot{W}_{cv} = 30.7 \text{ MW}$$

↑ Pump power required

$$\dot{W}_{total} = \dot{W}_A + \dot{W}_B = 391.8 \text{ kW} = 392 \text{ kW}$$

$$h_2 - h_3 = c_p (T_2 - T_3) = (1.004)(450 - 600) = -1.04 \text{ kJ/kg}$$

$$T_2 - T_3 = -150.6 \text{ K}$$

$$\dot{m} c_p (h_2 - h_3) = 0.6(-150.6) = -90.4 \text{ kW}$$

for h₂ - h₃: $h_2 - h_3 = c_p (T_2 - T_3) = 1.18(30 - 20) = 11.8 \text{ kJ/kg}$

$$-90.4 + \dot{m} c_p (h_2 - h_3) = 0$$

$$\dot{m} = \frac{90.4}{11.8} = 7.66 \text{ kg/s}$$

$$\dot{m} = \frac{\dot{W}_{cv}}{c_p (T_2 - T_3)} = \frac{10,000 \text{ kW}}{(1.004 \text{ kJ/kgK})(877 - 673 \text{ K})} = 24.0 \text{ kg/s}$$

$$\text{Heat exch: } \dot{m} c_p (T_2 - T_3) = \dot{m} c_p (T_4 - T_5)$$

$$T_2 = 673 \text{ K} + \frac{25(1.004)(1500 - 1200)}{(24.0)(206)} = 824 \text{ K}$$

7.18.87

$$\Delta T = 100^\circ - 20^\circ = 80^\circ \text{C}$$

$$\dot{Q}_H = \dot{m} c_p \Delta T = (0.5)(4100)(80) = 167,200 \text{ J}$$

$$t = \frac{\dot{Q}_{heat}}{\dot{Q}} = \frac{167,200}{850} = 196.7 \text{ s}$$

$$t_a = \frac{167,200}{850} = 196.7 \text{ s}$$

$$\dot{Q}_{cool} = \dot{m} h_{fg}$$

$$= (1.5)(2,257,000)$$

$$= 3,385,500 \text{ J}$$

$$t_b = \frac{\dot{Q}_{cool}}{\dot{Q}} = \frac{3,385,500}{850} = 3983 \text{ s}$$

$$h_1 \approx 244.5 \text{ kJ/kg}$$

$$h_2 \approx 280.2 \text{ kJ/kg} @ p = 10 \text{ kPa}$$

$$h_2 - h_1 = h_g - h_f \approx 35.7 \text{ kJ/kg}$$

$$a) \text{ Temp @ valve exit}$$

$$@ S.S. \dot{Q} = 0, \dot{W} = 0$$

$$\pm KE/PE$$

$$p_2 = 2 \text{ bar} \cdot 14.5 = 29 \text{ MPa}$$

$$T_{sat} = 29.46^\circ \text{C}$$

$$T_2 = (15 - 32) \frac{5}{9} = -9.4^\circ \text{C}$$

↑ interpolated value

$$h_5 - h_4 \approx c_p (T_5 - T_4) = 1.18(-10) = -11.8 \text{ kJ/kg}$$

$$\sum \dot{m} h_e - \sum \dot{m} h_i = 0$$

$$\dot{m} = \dot{m}_w \left(\frac{h_5 - h_4}{h_2 - h_1} \right) = \frac{-11.8}{35.7} = -0.33 \text{ kg/s}$$

$$= \frac{83.6}{9.15 \text{ kJ/kg}} = 9.1 \text{ kg/s}$$